

Week 5: The Flow of Electricity

Calculus Intuition: Slope, Area, Partial Derivatives, Jacobians, and
Gradients

Instructor / Mentor

Kiki Research Training Program

Week 5

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Review: Rotation as a Matrix

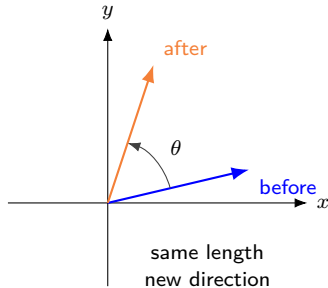
Last Class: A Matrix Can Move a Vector

Linear transformation review

A matrix is an operation. For a 2D vector,

$$\begin{bmatrix} x_{\text{new}} \\ y_{\text{new}} \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

- Rotation is one special linear transformation.
- It preserves length and changes direction.
- The four matrix entries come from sine and cosine.



Rotation Matrix: The Formula to Remember

Counterclockwise rotation by angle θ

$$R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}.$$

Quick check

If $\theta = 90^\circ$, then

$$R(90^\circ) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

So $[1, 0]^T$ becomes $[0, 1]^T$.

Common bug

Programming languages may use radians:

$$90^\circ = \frac{\pi}{2} \text{ radians.}$$

Interactive Warm-Up: Rotation Matrix Lab

Teacher setup

Use the commented HTML prompt in this `.tex` file to generate an interactive review page before class.

Student challenge:

1. Click *Generate Target Angle*.
2. Calculate the four matrix entries.
3. Enter them into the quiz.
4. Check whether the visual vector lands correctly.

What the visualization should show

- original vector
- correct target vector
- student's attempted vector
- angle arc and numeric error

Trigonometry as Projection and Slope

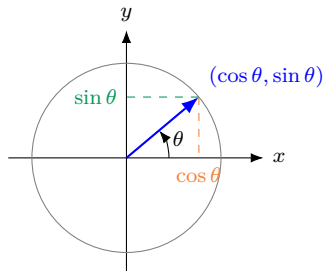
Why Sine and Cosine Appear in Rotation

Unit circle intuition

A point on the unit circle at angle θ has coordinates

$$(\cos \theta, \sin \theta).$$

- $\cos \theta$ measures horizontal projection.
- $\sin \theta$ measures vertical projection.
- Rotation is built by asking where the basis arrows go.



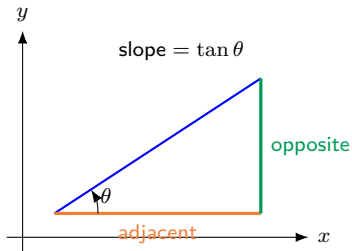
Tangent: The First Slope Story

Right-triangle meaning

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{\text{opposite}}{\text{adjacent}}.$$

Bridge to calculus

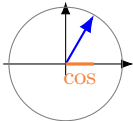
Tangent is already a slope idea. Calculus asks: what is the slope when the curve is not a straight line?



Three Trig Functions, Three Visual Meanings

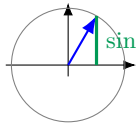
$$\cos \theta$$

Horizontal shadow of the unit arrow.



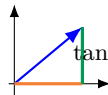
$$\sin \theta$$

Vertical shadow of the unit arrow.



$$\tan \theta$$

Slope of the ray from the origin.



Teacher move

Change the angle and ask students to predict which value increases, decreases, or changes sign.

Derivatives: Slope at an Instant

Question: What If the Line Becomes a Curve?

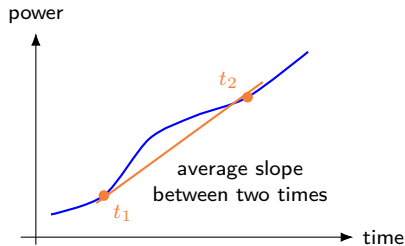
Straight line

One slope describes the whole line:

$$\text{slope} = \frac{\Delta y}{\Delta x}.$$

Curve

The slope changes from point to point. We need a local slope.



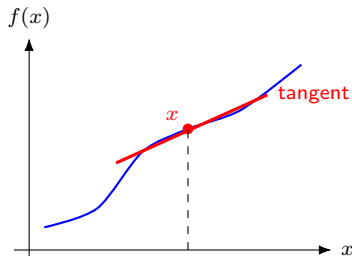
Derivative: Zoom In Until the Curve Looks Straight

Core idea

A derivative is the slope of the tangent line: the instantaneous rate of change.

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}.$$

- Large interval: average change.
- Tiny interval: local change.
- In grid data: how fast power demand is rising right now.



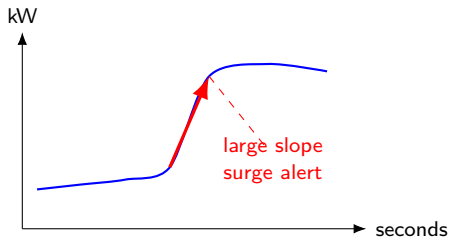
Electricity Example: Detecting a Power Surge

Sensor readings

If power readings are sampled every second,

$$\text{rate at second } k \approx P_{k+1} - P_k.$$

- A high positive derivative means demand is rising quickly.
- A high negative derivative means demand is dropping quickly.
- A surge is often a derivative signal, not just a high value.



Integrals: Accumulated Area

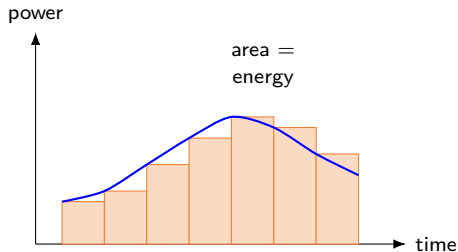
Integral: Area Means Total Accumulation

Core idea

If derivative is slope, integral is area:

$$\int_a^b P(t) dt = \text{total energy used.}$$

- Power is a rate: energy per time.
- Area under the power curve is energy.
- Smart meters approximate this by adding small rectangles.



Derivative and Integral Are Opposite Stories

Derivative

slope

How fast is the quantity changing now?

Grid question

Is power demand suddenly rising?

Big connection

Calculus translates between rates and totals.

Integral

area

How much has accumulated over time?

Grid question

How much energy was consumed today?

Multivariate Calculus: When Inputs Are Vectors

What If the Input Is Not One Number?

Scalar input

time $t \longrightarrow$ power $P(t)$.

One input direction means one derivative.

Vector input

$\begin{bmatrix} \text{temperature} \\ \text{hour of day} \\ \text{price} \\ \text{battery level} \end{bmatrix} \longrightarrow$ predicted demand.

Many input directions mean many rates of change.

Today's deeper question

If the input is a vector, how do we describe “rate of change”?

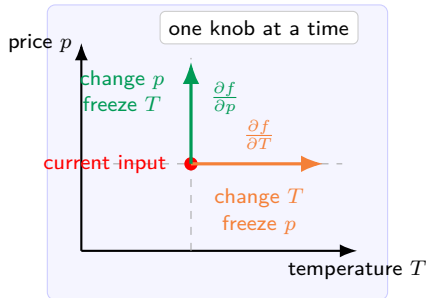
Partial Derivative: Change One Knob at a Time

Two-input function

Suppose demand depends on temperature and price:

$$D = f(T, p).$$

- $\frac{\partial f}{\partial T}$: change temperature, freeze price.
- $\frac{\partial f}{\partial p}$: change price, freeze temperature.
- Each partial derivative is a one-direction slope.



Jacobian Matrix: Many Outputs, Many Slopes

When a vector input produces a vector output

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{y} = \mathbf{f}(\mathbf{x}) = \begin{bmatrix} f_1(\mathbf{x}) \\ f_2(\mathbf{x}) \\ \vdots \\ f_m(\mathbf{x}) \end{bmatrix}.$$

The Jacobian collects all local rates of change

$$J(\mathbf{x}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}.$$

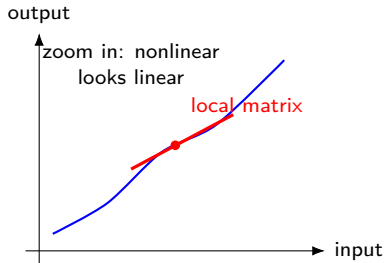
Jacobian Intuition: A Local Linear Transformation

Why Jacobian connects back to Week 4

Near one operating point, a nonlinear function can be approximated by a matrix:

$$\Delta \mathbf{y} \approx J(\mathbf{x}) \Delta \mathbf{x}.$$

- Matrix multiplication returns as a local approximation.
- The Jacobian says how small input changes become output changes.
- This is the bridge from linear algebra to AI optimization.



Gradient: Direction of Fastest Increase

Gradient: Put Partial Derivatives into One Vector

For a scalar output function

If $f(x, y)$ returns one number, its gradient is

$$\nabla f(x, y) = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}.$$

- It is a vector made of partial derivatives.
- It points toward fastest increase.
- Its length tells how fast the function increases there.

Grid interpretation

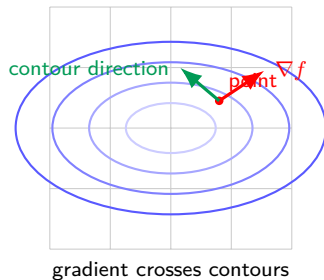
Which direction should we adjust price and battery dispatch to reduce overload fastest?

Visualizing the Gradient on a Contour Map

Contour map analogy

A contour line connects points with the same function value.

- Walking along a contour does not change height.
- The gradient points uphill most directly.
- Therefore the gradient is perpendicular to contour lines.



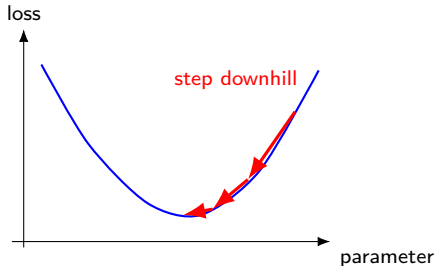
Gradient Descent: Why AI Cares About Calculus

Training a model

Machine learning often minimizes a loss function:

$$\text{new parameters} = \text{extoldparameters} - \eta \nabla L.$$

- The gradient tells the direction of steepest increase.
- Negative gradient points downhill.
- This is how models learn from errors.



Class Flow and Homework

Today's Storyline

1. **Rotation review:** matrix multiplication moves vectors.
2. **Trig review:** sine/cosine are projections; tangent is slope.
3. **Derivative:** slope of a curve at one instant.
4. **Integral:** area under a rate curve gives total accumulation.
5. **Partial derivative:** one input direction at a time.
6. **Jacobian and gradient:** rate of change for vector inputs and AI models.

Key mindset

Calculus is not only formula memorization. It is a language for asking: *if I change this input a little, what happens next?*

Vibe Coding Task: Power Surge Detector

Prompting hints for your AI assistant

1. Ask for a Python script that creates noisy second-by-second power readings.
2. Ask it to calculate discrete slopes using consecutive differences.
3. Ask it to mark time points where the slope exceeds a threshold.
4. Ask it to plot power and derivative on two aligned charts.
5. Ask it to explain why a surge can be detected by rate of change.

Student responsibility

Do not trust the plot blindly. Check units: if power is kW and time is seconds, the derivative is kW per second.

Homework for This Week

1. Given second-by-second power data, complete the loop that calculates the slope between consecutive seconds.
2. Mark all time points where the slope is larger than a chosen surge threshold.
3. Write one paragraph explaining derivative as slope and integral as area using the power-consumption example.
4. Bonus: for $f(T, p)$, explain in plain English the difference between $\partial f / \partial T$ and $\partial f / \partial p$.

Thank you!

Next week: reading papers deeply and developing research taste.